Study Design

Purpose: State the paper’s objective and the full analytical workflow.

Key sentences:

- We evaluated how hierarchical simplification of multichannel river networks alters network geometry and event-scale routing response.

- The workflow combined real-case selection, mask-based network extraction, hierarchy detection, mask-based collapse and regeneration, hydraulic attribute reconstruction, equivalent-resistance analysis, and comparative routing simulations.

Include:

- real cases and synthetic support experiments

- geometry and topology metrics

- equivalent resistance

- event-scale routing comparison

Case Selection

Purpose: Explain why the examples were chosen and how they were constrained.

Key sentences:

- Study cases were defined as multichannel river sections bounded by upstream and downstream gauging stations.

- Cases were retained only where geomorphic complexity and hydrologic data availability supported event-scale comparative analysis.

Include:

- gauge-pair logic

- SWORD matching

- subdaily discharge availability

- event suitability and quality screening

- final retained study set

Figure suggestion: map of selected cases.

Synthetic Support Experiments

Purpose: Present synthetic networks as a methodological support component.

Key sentences:

- Synthetic networks were used as controlled methodological support to isolate the effects of topology, width partitioning, slope contrasts, sinuosity contrasts, and forcing characteristics.

- These experiments supported interpretation of the real-network simplification results rather than replacing them.

Include:

- base corridor setup

- looped and cross-linked perturbations

- width-conserving branch partitioning

- sensitivity dimensions

Figure suggestion: schematic of the synthetic setup.

Mask Preparation and RivGraph Extraction

Purpose: Describe the image-to-network conversion.

Key sentences:

- Source water masks were converted to binary rasters and manually cleaned where necessary before centerline and network extraction with RivGraph.

- This produced centerline, node, and link geometries for each case.

Include:

- binary raster preparation

- manual mask cleaning

- RivGraph extraction

Directed Graph Preparation

Purpose: Explain how direction assignment was handled and frame manual review as optional.

Key sentences:

- Flow direction was assigned primarily from RivGraph outputs.

- Where automated direction assignment was uncertain, manual review of link direction and inlet–outlet roles could be used as an additional quality-control step.

- All downstream analyses required a valid directed network with one source and one sink.

Include:

- RivGraph flow assignment

- optional manual review and correction

- graph validation criteria

Hierarchical Unit Detection

Purpose: Define the core analysis object.

Key sentences:

- The network was decomposed into bifurcation–confluence units, each defined by one or more directed paths between an upstream bifurcation and a downstream confluence.

- Units could be simple, multi-thread, or compound and nested.

Include:

- path-based unit definition

- nested units

- compound bubbles as broader multichannel zones

Figure suggestion: annotated unit example on a real network.

Metrics and Collapse Ranking

Purpose: Justify how candidate collapses were prioritized.

Key sentences:

- Candidate units were characterized using geometric persistence, path-width disparity, and internal topologic complexity.

- Collapse ranking was based on path disparity, elongation, equivalent length, and topologic complexity.

Include:

- path_disparity_width

- equivalent_length

- elongation

- topologic_complexity_score

- grouped candidates selected from ranked contiguous partitions

Collapse and Variant Regeneration

Purpose: Explain that simplification was implemented in mask space and regenerated with RivGraph.

Key sentences:

- Simplification was implemented by modifying the parent mask within selected collapse domains and rerunning RivGraph, rather than deleting graph elements directly.

- This ensured that each simplified state was a regenerated network geometry.

Include:

- footprint construction

- filling enclosed dry holes

- independent and sequential experiments

- realized versus anchored interpretations if used in the paper

Figure suggestion: original versus collapsed example.

Width Reconstruction

Purpose: Describe the width methodology clearly and concisely.

Key sentences:

- Widths were recomputed on regenerated variants using a two-bank distance-transform-based method.

- For each regenerated link, total width was measured on the collapsed mask, wet width on the original wet mask, and dry width as the difference between the two.

Include:

- sample-based bank search

- total, wet, and dry width families

- why wet versus dry width matters after collapse

Figure suggestion: width-method figure, ideally combined with original and collapsed geometry.

Hydraulic Context Reconstruction

Purpose: Explain how slope and profile context were reconstructed.

Key sentences:

- Node water-surface-elevation context from matched external reference nodes was converted to a monotone or interpolated longitudinal profile for slope estimation.

- Observed node elevation context and derived routing profile were treated as distinct quantities.

Include:

- SWORD node matching

- profile-based slope derivation

Geometry-Informed Equivalent Resistance

Purpose: Position equivalent resistance as a core analytical layer.

Key sentences:

- Equivalent resistance was used as a geometry-informed predictor of the hydraulic effect of multichannel structures and their simplified replacements.

- Edge-level resistances were computed from link geometry and reduced to equivalent corridor resistance under normalized head boundary

conditions.

- The principal experiment-level comparison metric was the ratio between realized child-corridor resistance and matched base-corridor resistance.

Include:

- state-level bubble resistance

- experiment-level matched comparison

- interpretation of Lambda_matched

Figure suggestion: schematic of bubble reduction and matched corridor comparison.

Event Forcing and Routing Comparison

Purpose: Explain the comparative hydrodynamic experiment.

Key sentences:

- All states from a given example were forced with the same observed event hydrograph and routed with the same parameterization.

- Results were interpreted as relative differences in timing, attenuation, and hydrograph shape across variants.

Include:

- shared forcing per example

- shared routing parameterization

- comparative rather than state-specific calibration framing

- peak timing, attenuation, rise time, fall time, and hydrograph shape

Suggested Figure Set

- Figure 1: map of selected real cases

- Figure 2: synthetic network setup and sensitivity dimensions

- Figure 3: hierarchy definition on a real reviewed network

- Figure 4: collapse and regeneration workflow on one example

- Figure 5: width methodology on original versus collapsed geometry

- Figure 6: equivalent-resistance schematic and matched comparison concept